

OpenCV Analysis: Strong Coupling ↔ Weak Coupling Transition in Laser Honeycomb Setup

The video shows a dynamic laser projection experiment using honeycomb serpentine privacy film, polarizing films, and N20 motor rotation. GPTs interpretation of strong-weak coupling transition is highly appropriate.

Metric	Value
Number of Dots	0 to 11 (average 3.8)
Spread (cluster dispersion)	0 to 130+ pixels
Observed Phases	Strong Coupling (tight cluster) to Weak Coupling (satellite formation)
Transition Period	Approximately 1-2 seconds driven by motor rotation

Physical Interpretation:

- Honeycomb serpentine film acts as a periodic spatial filter.
- Polarizing films (diamond + square orientation) provide polarization-dependent interference.
- Motor rotation sweeps the phase/coupling parameter.

This setup optically analogizes strong-weak coupling transitions seen in cavity QED, moiré physics, or coupled oscillator systems. As rotation changes the effective coupling strength, the laser dots switch between collective (coherent cluster) and individual (dispersed satellite) modes.

OpenCV blob detection confirms clear bimodal transitions with significant variance in dot count and spread, matching the strong ↔ weak coupling description.